
OPENSr-SRGAN: A FLEXIBLE SUPER-RESOLUTION FRAMEWORK FOR MULTISPECTRAL EARTH OBSERVATION DATA *

Simon Donike[✉], Cesar Aybar[✉], Julio Contreras[✉], Luis Gómez-Chova[✉]

Image and Signal Processing Group

University of Valencia

Valencia, Spain

simon.donike@uv.es

ABSTRACT

We present *OpenSR-SRGAN*, an open and modular framework for single-image super-resolution in Earth Observation. The software provides a unified implementation of SRGAN-style models that is easy to configure, extend, and apply to multispectral satellite data such as Sentinel-2. Instead of requiring users to modify model code, *OpenSR-SRGAN* exposes generators, discriminators, loss functions, and training schedules through concise configuration files, making it straightforward to switch between architectures, scale factors, and band setups. The framework is designed as a practical tool and benchmark implementation rather than a state-of-the-art model. It ships with ready-to-use configurations for common remote sensing scenarios, sensible default settings for adversarial training, and built-in hooks for logging, validation, and large-scene inference. By turning GAN-based super-resolution into a configuration-driven workflow, *OpenSR-SRGAN* lowers the entry barrier for researchers and practitioners who wish to experiment with SRGANs, compare models in a reproducible way, and deploy super-resolution pipelines across diverse Earth-observation datasets.

1 Introduction

Optical satellite imagery plays a key role in monitoring the Earth’s surface for applications such as agriculture [1], land cover mapping [2], ecosystem assessment [3], and disaster management [4]. The European Space Agency’s Sentinel-2 mission provides freely available multispectral imagery at 10 m spatial resolution with a revisit time of five days, enabling dense temporal monitoring at global scale. In contrast, very-high-resolution sensors, such as Pleiades or SPOT, offer much richer spatial detail but limited temporal coverage and high acquisition costs. Consequently, a trade-off exists between spatial and temporal resolution in Earth-Observation (EO) imagery.

Single-image super-resolution (SISR) aims to enhance the spatial detail of low-resolution (LR) observations by learning a mapping to a plausible high-resolution (HR) counterpart. In remote sensing, SR can bridge the gap between freely available medium-resolution imagery and costly commercial data, potentially improving downstream tasks such as land-cover classification, object detection, and change detection. The advent of deep convolutional networks led to major breakthroughs in both reconstruction fidelity and perceptual realism [5, 6].

Generative Adversarial Networks (GANs) [7] introduced an adversarial learning framework in which a generator and a discriminator are trained in competition, enabling the synthesis of realistic, high-frequency image details. Since their introduction, GANs have been rapidly adopted in the remote sensing community for tasks such as cloud removal, image translation, domain adaptation, and data synthesis [8, 9]. These applications demonstrated the potential of adversarial training to generate spatially coherent and perceptually plausible remote sensing imagery.

Building on these successes, the computer-vision community introduced the Super-Resolution GAN (SRGAN) [10], which combined perceptual and adversarial losses to reconstruct photo-realistic high-resolution images from their low-resolution counterparts. The approach inspired a wave of research applying SRGAN-like architectures to remote

*Published as Software Package <https://github.com/ESAOpenSR/SRGAN>, DOI: 10.5281/zenodo.17590993

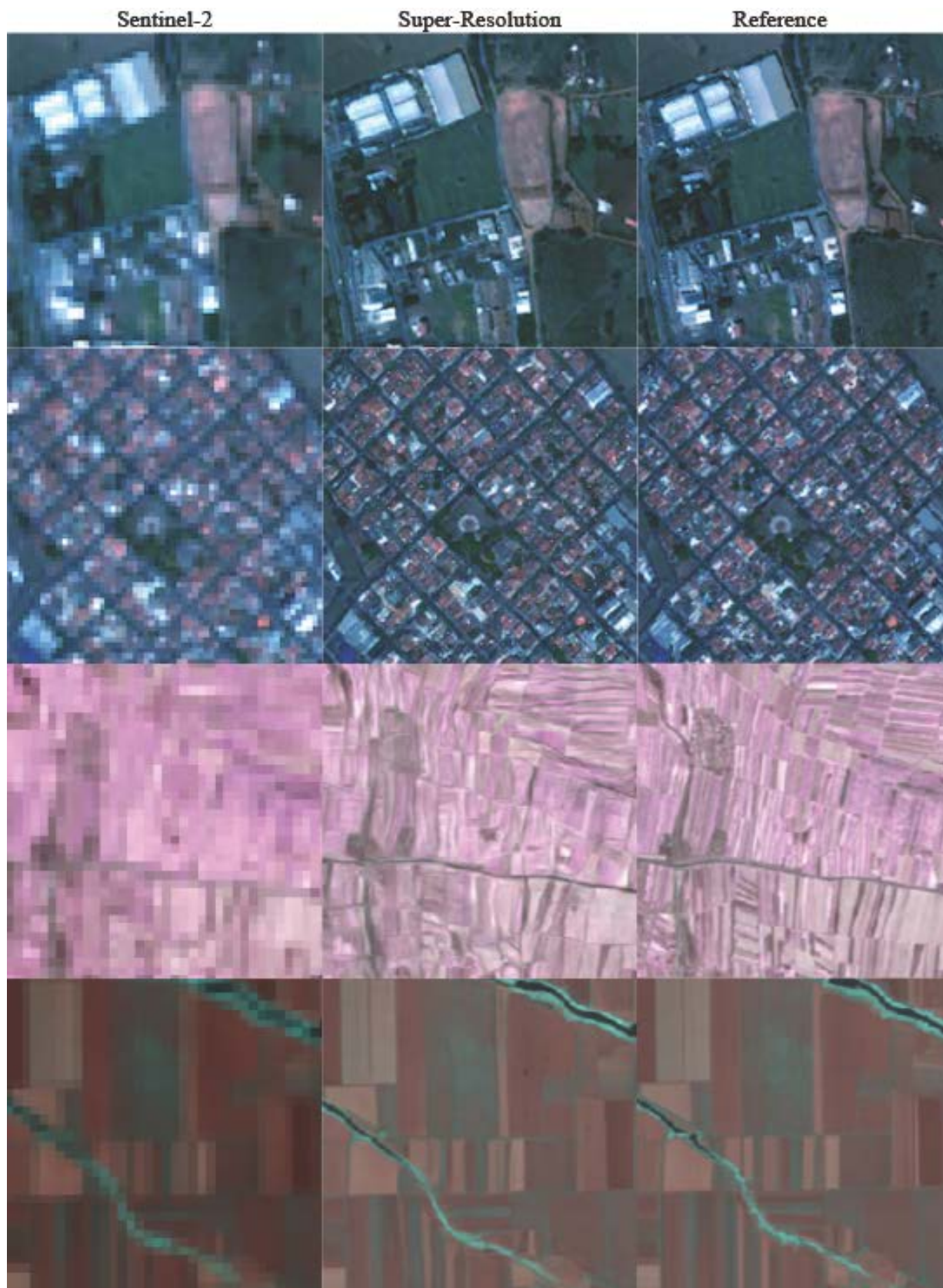


Figure 1: Sentinel-2 Super-Resolution examples: $4\times$ RGB (top two rows), $8\times$ SWIR (bottom two rows)

sensing super-resolution [11, 12, 13, 14], where the ability to recover fine spatial detail from coarse observations can significantly enhance analysis of land cover, infrastructure, and environmental change.

Recent advances in diffusion and transformer-based architectures have shifted the state of the art in image super-resolution toward generative models with stronger probabilistic and contextual reasoning [15, 16, 17]. Nevertheless, GAN-based approaches continue to be actively explored [18] and remain a practical choice for operational production settings [19].

2 Statement of Need

Despite their success in computer vision, Generative Adversarial Networks (GANs) remain notoriously difficult to train [20, 21, 22]. The simultaneous optimization of generator and discriminator networks often leads to unstable dynamics, mode collapse, and high sensitivity to hyperparameters.

In remote sensing applications, these issues are amplified by domain-specific challenges such as multispectral or hyperspectral inputs, high dynamic range reflectance values, varying sensor characteristics, and limited availability of perfectly aligned high-resolution ground-truth data.

Moreover, researchers in remote sensing rarely work with fixed RGB imagery. They frequently need to adapt existing GAN architectures to support arbitrary numbers of spectral bands, retrain models for different satellite sensors (e.g., Sentinel-2, SPOT, Pleiades, PlanetScope), or implement benchmarks for newly collected datasets. These modifications usually require non-trivial changes to the model architecture, preprocessing pipeline, and loss configuration, making reproducibility and experimentation cumbersome.

Implementing the full set of heuristics that make GAN training stable, such as generator pretraining, adversarial loss ramping, label smoothing, learning-rate warmup, and exponential moving-average (EMA) tracking, adds another layer of complexity. Consequently, reproducing and extending GAN-based SR models in the Earth-Observation (EO) domain is often time-consuming, fragile, and inconsistent across studies.

3 Contribution Summary

OpenSR-SRGAN was developed to address these challenges by providing a unified, modular, and extensible framework for training and evaluating GAN-based super-resolution models in remote sensing. The software integrates multiple state-of-the-art SR architectures, loss functions, and training strategies within a configuration-driven design that allows users to flexibly adapt experiments without modifying the source code.

The main contributions of this work include:

- **Modular GAN framework:** Supports interchangeable generator and discriminator architectures with customizable depth, width, and scale factors.
- **Configuration-first workflow:** Enables fully reproducible training and evaluation through concise YAML files, independent of code changes.
- **Training stabilization techniques:** Includes generator pretraining, adversarial ramp-up, learning-rate warmup, label smoothing, and EMA smoothing.
- **Multispectral compatibility:** Provides native support for arbitrary band configurations from different satellite sensors.
- **OpenSR ecosystem integration:** Connects seamlessly to the SEN2NAIP dataset, leverages the unified evaluation framework `opensr-test` [23], and supports scalable inference via `opensr-utils` [24].

Together, these features make *OpenSR-SRGAN* a reliable boilerplate for researchers and practitioners to train, benchmark, and deploy GAN-based SR models across diverse Earth-observation datasets and sensor types.

4 Software Overview and Framework Design

OpenSR-SRGAN follows a modular and configuration-driven architecture. All model definitions, loss compositions, and training schedules are controlled through a single YAML configuration file, ensuring that experiments remain reproducible and easily adaptable to new sensors, band configurations, or datasets. The framework is implemented in PyTorch and PyTorch Lightning, providing seamless GPU acceleration and built-in experiment logging.

The system consists of four main components:

- A flexible **generator–discriminator architecture** supporting multiple SR backbones.
- A configurable **multi-loss system** combining pixel, perceptual, spectral, and adversarial objectives.
- A robust **training pipeline** with pretraining, warmup, ramp-up, and EMA stabilization mechanisms.
- Integration with the **OpenSR ecosystem** for dataset access, evaluation (`opensr-test`), and large-scale inference (`opensr-utils`).

Implementation details of the available generator and discriminator variants, training features, and loss components are summarised in Appendix A. A detailed list of the internal diagnostics logged during training is provided in Appendix B.

4.1 Generator Architectures

The generator network can be configured with different backbone types, each providing a unique trade-off between complexity, receptive field, and textural detail (see Table 1 in Appendix A).

The `Generator` class provides a unified implementation of SR backbones that share a common convolutional structure while differing in their internal residual block design. The module is initialized with a `model_type` flag selecting one of `{res, rcab, rrdb, lka, esrgan, cgan}`, each drawn from a shared registry of block factories or dedicated ESRGAN implementation. Given an input tensor x , the model applies a wide receptive-field head convolution, followed by N residual blocks of the selected type, a tail convolution for residual fusion, and an upsampling module that increases spatial resolution by a factor of 2, 4, or 8. The network ends with a linear output head producing the super-resolved image:

$$x' = \text{Upsample}(\text{Conv}_{\text{tail}}(\text{Body}(x_{\text{head}}) + x_{\text{head}})). \quad (1)$$

This modular structure allows researchers to experiment with different block designs—standard residual, channel attention (RCAB), dense residual (RRDB), large-kernel attention (LKA), or noise-augmented variants—without altering the training pipeline or configuration schema. All models share identical input-output interfaces and residual scaling for stability, ensuring drop-in interchangeability across experiments.

4.1.1 Conditional Generator with Noise Injection

The `cgan` variant extends the standard generator by conditioning on both the low-resolution image and a latent noise vector z in order to enhance the sampling diversity. It replaces standard residual blocks with `NoiseResBlocks`, which introduce controlled stochasticity. Each `NoiseResBlock` uses a small Multi-Layer Perceptron (MLP) to project the noise code $z \in \mathbb{R}^d$ into per-channel scale and bias parameters (γ, β) that modulate intermediate activations before the nonlinearity:

$$x_{\text{mod}} = (1 + \gamma) \odot \text{Conv}_1(x) + \beta \quad (2)$$

followed by a second convolution and residual fusion:

$$x' = x + \text{Conv}(\text{PReLU}(x_{\text{mod}})) \cdot s \quad (3)$$

where s is a residual scaling factor. This mechanism maintains spatial coherence while enabling the generation of multiple plausible high-frequency realizations for the same LR input. During training, a random noise vector is sampled per image; during inference, users may supply explicit latent codes or fix random seeds for deterministic behavior. The module also exposes `sample_noise(batch_size)` and a `return_noise` flag for reproducibility and logging.

4.2 Discriminator Architectures

The discriminator can be selected to prioritize either global consistency or fine local realism. The different architectures and their purposes are outlined in Table 2 in Appendix A. Three discriminator variants are implemented to complement the different generator types: a global `Discriminator`, a local `PatchGANDiscriminator`, and the deeper `ESRGANDiscriminator`. All are built from shared convolutional blocks with LeakyReLU activations and instance normalization.

The standard discriminator follows the original SRGAN [10] design and evaluates the realism of the entire super-resolved image and the actual HR image. It stacks a sequence of strided convolutional layers with progressively increasing feature channels, an adaptive average pooling layer to a fixed spatial size, and two fully connected layers producing a scalar real/fake score. This “global” discriminator promotes coherent large-scale structure and overall photorealism.

The PatchGANDiscriminator instead outputs a grid of patch-level predictions, classifying each overlapping region as real or fake. Built upon the CycleGAN/pix2pix [25, 26] reference implementation, it uses a configurable number of convolutional layers and normalization schemes (batch, instance, or none). The resulting patch map acts as a spatial realism prior, emphasizing texture fidelity and fine detail.

Finally, the ESRGANDiscriminator mirrors the deeper VGG-style stack from ESRGAN. Its `base_channels` and fully connected `linear_size` can be tuned to match the generator capacity, offering an aggressive adversarial signal when paired with RRDB-based generators.

Together, these architectures allow users to select the appropriate adversarial granularity: global consistency through SRGAN-style discrimination, local realism through PatchGAN, or sharper perceptual contrast via ESRGAN.

4.3 Training Features

Training stability is improved through several built-in mechanisms that address common issues of adversarial optimization (summarized in Table 3, Appendix A). These are configured in the Training section of the YAML config file.

4.3.1 General Training Optimizations

Several additional methods contribute to stable adversarial optimization. Label smoothing replaces hard discriminator targets (1 for real, 0 for fake) with softened values such as 0.9 and 0.1, preventing overconfidence and promoting smoother gradients. A short generator warmup phase allows G to learn basic low-frequency structure before adversarial feedback is introduced, often combined with a linear or cosine learning-rate ramp to avoid abrupt updates. The discriminator holdback delays D updates for the first few epochs so that G can stabilise; when enabled, D also follows a short warmup schedule to balance learning rates. Finally, both optimisers employ adaptive scheduling via ReduceLROnPlateau, lowering the learning rate when progress stagnates. These implementations mitigate divergence and improve convergence stability in adversarial training. All of these techniques can be configured from the config file as the unified entry-point.

4.3.2 Exponential Moving Average (EMA) Stabilisation

The EMA mechanism [27] is an optional stabilisation technique applied to the generator weights to produce smoother outputs and more reliable validation metrics, commonly used throughout model training pipelines in general and generative image applications in particular. Instead of evaluating the generator using its raw, rapidly fluctuating parameters, an auxiliary set of duplicate weights θ_{EMA} is maintained as a smoothed version of the online weights θ . After each training step, the model parameters are updated as an exponential moving average:

$$\theta_{\text{EMA}}^{(t)} = \beta \theta_{\text{EMA}}^{(t-1)} + (1 - \beta) \theta^{(t)} \quad (4)$$

where $\beta \in [0, 1)$ is the decay factor controlling how much past states influence the smoothed estimate. A higher β (e.g., 0.999) gives longer memory and stronger smoothing, while a lower value responds more quickly to new updates.

During validation and inference, the EMA parameters replace the instantaneous generator weights, yielding more temporally consistent reconstructions and reducing the variance of perceptual and adversarial loss curves. The inference process thus evaluates:

$$\hat{y}_{\text{SR}} = G(x; \theta_{\text{EMA}}) \quad (5)$$

where $\hat{y}_{\text{SR}}$ denotes the final super-resolved output produced by the EMA-stabilised generator. Empirically, applying EMA has been shown to stabilise adversarial training by mitigating oscillations between the generator and discriminator and by improving the perceptual smoothness and reproducibility of the resulting super-resolved images [28].

4.4 Loss Functions

Each loss term (see Table 4 in Appendix A) can be weighted independently, allowing users to balance spectral accuracy and perceptual realism.

Typical configurations combine L1, Perceptual, and Adversarial losses, optionally augmented by SAM and TV for multispectral consistency and smoothness. The overall objective is a weighted sum of these terms defined in the `Training.Losses` section of the configuration.

A detailed description of the internal training and validation metrics logged alongside these losses is given in Appendix B.

5 Limitations

Super-resolution techniques, including those implemented in *OpenSR-SRGAN*, can enhance apparent spatial detail but can never substitute for true high-resolution observations acquired by native sensors. While *OpenSR-SRGAN* provides a stable and extensible foundation for GAN-based super-resolution in remote sensing, several limitations remain. First, the framework focuses on the engineering and reproducibility aspects of model development rather than achieving state-of-the-art quantitative performance. It is therefore intended as a research and benchmarking blueprint, not as an optimized production model. Second, although the modular configuration system greatly simplifies experimentation, users are still responsible for ensuring proper data preprocessing, radiometric normalization, and accurate LR–HR alignment, factors that strongly influence training stability and reconstruction quality. Third, adversarial optimization in multispectral domains remains sensitive to dataset size and diversity; small or unbalanced datasets may still yield mode collapse or spectral inconsistencies despite the provided stabilization mechanisms. Finally, the current release does not include native uncertainty estimation or automatic hyperparameter tuning; these remain open areas for future extension.

Acknowledgement

This work has been supported by the European Space Agency (ESA) Φ -Lab, within the framework of the ‘Explainable AI: Application to Trustworthy Super-Resolution (OpenSR)’ Project.

References

- [1] M. Weiss, F. Jacob, and G. Duveiller. Remote sensing for agricultural applications: A meta-review. *Remote Sensing of Environment*, 236:111402, 2020.
- [2] Jwan Al-Doski, Shattri B Mansor, H Ng, P San, and Z Khuzaimah. Land cover mapping using remote sensing data. *American Journal for Geographical Information Systems*, 2020:33–45, 2020.
- [3] Zhaoqin Li, Dandan Xu, and Xulin Guo. Remote sensing of ecosystem health: Opportunities, challenges, and future perspectives. *Sensors*, 14(11):21117–21139, 2014.
- [4] Rami Ahmad. Smart remote sensing network for disaster management: An overview. *Telecommunication Systems*, 87(1):213–237, 2024.
- [5] Chao Dong, Chen Change Loy, Kaiming He, and Xiaoou Tang. Image super-resolution using deep convolutional networks, 2015.
- [6] Jiwon Kim, Jung Kwon Lee, and Kyoung Mu Lee. Deeply-recursive convolutional network for image super-resolution, 2016.
- [7] Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial networks, 2014.
- [8] Simon Donike, Cesar Aybar, Julio Contreras, and Luis Gómez-Chova. Near-infrared band synthesis from earth observation imagery with learned location embeddings and task-driven loss functions. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 18:24319–24330, 2025.
- [9] Xingzhe Su, Wenwen Qiang, Jie Hu, Fengge Wu, Changwen Zheng, and Fuchun Sun. Intriguing property and counterfactual explanation of gan for remote sensing image generation, 2024.
- [10] Christian Ledig et al. Photo-realistic single image super-resolution using a generative adversarial network. In *CVPR*, 2017.
- [11] Xuan Wang, Lijun Sun, Abdellah Chehri, and Yongchao Song. A review of gan-based super-resolution reconstruction for optical remote sensing images. *Remote Sensing*, 15(20), 2023.
- [12] Sen Jia, Zhihao Wang, Qingquan Li, Xiuping Jia, and Meng Xu. Multiattention generative adversarial network for remote sensing image super-resolution. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–15, 2022.
- [13] Fanen Meng, Sensen Wu, Yadong Li, Zhe Zhang, Tian Feng, Renyi Liu, and Zhenhong Du. Single remote sensing image super-resolution via a generative adversarial network with stratified dense sampling and chain training. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–22, 2024.
- [14] Piper Wolters, Favyen Bastani, and Aniruddha Kembhavi. Zooming out on zooming in: Advancing super-resolution for remote sensing, 2023.

- [15] Yi Xiao, Qiangqiang Yuan, Kui Jiang, Jiang He, Xianyu Jin, and Liangpei Zhang. Ediffrs: An efficient diffusion probabilistic model for remote sensing image super-resolution. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–14, 2024.
- [16] Jinzhe Liu, Zhiqiang Yuan, Zhaoying Pan, Yiqun Fu, Li Liu, and Bin Lu. Diffusion model with detail complement for super-resolution of remote sensing. *Remote Sensing*, 14(19), 2022.
- [17] Simon Donike, Cesar Aybar, Luis Gómez-Chova, and Freddie Kalaitzis. Trustworthy super-resolution of multi-spectral sentinel-2 imagery with latent diffusion. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 18:6940–6952, 2025.
- [18] Daniel Panangian and Ksenia Bittner. Can location embeddings enhance super-resolution of satellite imagery?, 2025.
- [19] Piper Wolters, Favien Bastani, and Aniruddha Kembhavi. Zooming out on zooming in: Advancing super-resolution for remote sensing, 2023.
- [20] David Bau, Jun-Yan Zhu, Jonas Wulff, William Peebles, Hendrik Strobelt, Bolei Zhou, and Antonio Torralba. Seeing what a gan cannot generate. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, October 2019.
- [21] Tim Salimans, Ian Goodfellow, Wojciech Zaremba, Vicki Cheung, Alec Radford, Xi Chen, and Xi Chen. Improved techniques for training gans. In D. Lee, M. Sugiyama, U. Luxburg, I. Guyon, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 29. Curran Associates, Inc., 2016.
- [22] Christopher Bowles, Liang Chen, Ricardo Guerrero, Paul Bentley, Roger Gunn, Alexander Hammers, David Alexander Dickie, Maria Valdés Hernández, Joanna Wardlaw, and Daniel Rueckert. Gan augmentation: Augmenting training data using generative adversarial networks, 2018.
- [23] Cesar Aybar, David Montero, Simon Donike, Freddie Kalaitzis, and Luis Gómez-Chova. A comprehensive benchmark for optical remote sensing image super-resolution. *IEEE Geoscience and Remote Sensing Letters*, 21:1–5, 2024.
- [24] Simon Donike and contributors. Opensr-utils: Large-scale inference and data processing utilities for super-resolution of remote sensing imagery. <https://github.com/ESA0penSR/opensr-utils>, 2025. Accessed: 2025-10-14.
- [25] Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A. Efros. Unpaired image-to-image translation using cycle-consistent adversarial networks, 2020.
- [26] Phillip Isola, Jun-Yan Zhu, Tinghui Zhou, and Alexei A. Efros. Image-to-image translation with conditional adversarial networks, 2018.
- [27] Daniel Morales-Brotons, Thijs Vogels, and Hadrien Hendrikx. Exponential moving average of weights in deep learning: Dynamics and benefits, 2024.
- [28] Xiuliang Xi, Xin Jin, Qian Jiang, Yu Lin, Wei Zhou, and Lei Guo. Ema-gan: A generative adversarial network for infrared and visible image fusion with multiscale attention network and expectation maximization algorithm. *Advanced Intelligent Systems*, 5(11):2300310, 2023.
- [29] Jie Cai, Zibo Meng, and Chiu Man Ho. Residual channel attention generative adversarial network for image super-resolution and noise reduction, 2020.
- [30] Yulun Zhang, Yapeng Tian, Yu Kong, Bineng Zhong, and Yun Fu. Residual dense network for image super-resolution, 2018.
- [31] Liam Chalcraft, Ruben Lourenço Pereira, Mikael Brudfors, Andrew S. Kayser, Mark D’Esposito, Cathy J. Price, Ioannis Pappas, and John Ashburner. Large-kernel attention for efficient and robust brain lesion segmentation, 2023.
- [32] Ugur Demir and Gozde Unal. Patch-based image inpainting with generative adversarial networks, 2018.
- [33] Cesar Aybar, David Montero, Julio Contreras, Simon Donike, Freddie Kalaitzis, and Luis Gómez-Chova. Sen2naip: A large-scale dataset for sentinel-2 image super-resolution. *Scientific Data*, 11(1):1389, 2024.

Appendix A Architecture and Training Components

Table 1: **Implemented generator types and their characteristics.**

Generator Type	Description
res [10]	SRResNet generator using residual blocks without batch normalization. Stable and effective for content pretraining.
rcab [29]	Residual Channel Attention Blocks. Adds channel-wise reweighting to enhance textures and small structures.
rrdb [30]	Residual-in-Residual Dense Blocks (RRDB) as in ESRGAN. Deep structure with dense connections, improving detail sharpness.
lka [31]	Large-Kernel Attention blocks. Capture wide spatial context, beneficial for structured RS patterns (e.g., fields, roads).
esrgan [30]	Full ESRGAN generator with configurable RRDB count, growth channels, and residual scaling.
cgan	Stochastic Conditional Generator with NoiseResBlock.

Table 2: **Implemented discriminator types and their purposes.**

Discriminator Type	Description
standard [10]	A global SR-GAN-style CNN discriminator that judges the overall realism of the full image. Promotes coherent global structure.
patchgan [32]	A PatchGAN discriminator that outputs patch-level predictions. Focuses on local realism and texture detail. Patch size is implicitly controlled by network depth (<code>n_blocks</code>).
esrgan [30]	ESRGAN discriminator with configurable base channels and linear head size to complement RRDB generators.

Table 3: **Implemented training features for stable adversarial optimization.**

Feature	Description
<code>pretrain_g_only</code>	Trains only the generator (content losses) for a specified number of steps (<code>g_pretrain_steps</code>) before enabling the adversarial loss.
<code>adv_loss_ramp_steps</code>	Gradually increases the weight of the adversarial loss from 0 to the maximum value (<code>adv_loss_beta</code>), improving stability.
<code>label_smoothing</code>	Applies soft labels (e.g., 0.9 for real) to stabilize the discriminator and reduce overconfidence.
<code>g_warmup_steps</code> , <code>g_warmup_type</code> <code>EMA.enabled</code>	Warmup schedule for the generator’s learning rate, linear or cosine, ensuring smooth optimizer convergence. Enables Exponential Moving Average tracking of generator weights for smoother validation and inference outputs.
TTUR LRs (<code>optim_g_lr</code> , <code>optim_d_lr</code>) Adam betas/epsilon (<code>betas</code> , <code>eps</code>)	Two-time-scale update rule with discriminator LR defaulting to a slower schedule than the generator to maintain balance. GAN-friendly defaults (0.0, 0.99) and 10^{-7} avoid stale momentum and numerical noise during adversarial updates.
Weight-decay exclusions	Normalisation and bias parameters are automatically removed from decay groups so regularisation targets convolutional kernels only.
Plateau scheduler (<code>cooldown</code> , <code>min_lr</code>) <code>gradient_clip_val</code>	ReduceLROnPlateau schedulers for G and D now support cooldown periods and minimum learning-rate floors. Optional global-norm clipping applied after every optimiser step to suppress discriminator-induced spikes.
<code>Training.gpus</code>	Enables distributed data-parallel training when multiple GPU indices are listed, scaling training efficiently via Pytorch-Lightning.

Table 4: **Supported loss components and configuration parameters.**

Loss Type	Description
L1 Loss	Pixel-wise reconstruction loss using the L1 norm; maintains global content and brightness consistency.
SAM Loss	Spectral Angle Mapper; penalizes angular differences between spectral vectors of predicted and true pixels, preserving spectral fidelity.
Perceptual Loss	Feature-space loss using pre-trained VGG19 or LPIPS metrics; improves perceptual quality and texture realism.
TV Loss	Total Variation regularizer; encourages spatial smoothness and reduces noise or artifacts.
Adversarial Loss	Binary cross-entropy loss on discriminator predictions; drives realism and high-frequency texture generation.

Appendix B Internal Metrics

During training, a range of scalar metrics are continuously computed and logged in Weights & Biases. These indicators capture the evolving balance between generator and discriminator, quantify loss dynamics, and provide early warnings of instability or mode collapse. Together, they form a compact diagnostic suite that allows users to monitor convergence, identify regime transitions (e.g., from pretraining to adversarial learning), and ensure stable training behaviour. Table 5 summarises the most relevant internal metrics recorded by *OpenSR-SRGAN*.

Table 5: **Key internal metrics tracked during training and validation for monitoring adversarial dynamics, generator stability, and EMA behaviour.**

Metric	Description and Expected Behaviour
training/pretrain_phase	Binary flag indicating whether generator-only warm-up is active. Remains 1 during pretraining and switches to 0 once adversarial learning begins.
discriminator/adversarial_loss	Binary cross-entropy loss separating real HR from generated SR samples. Decreases below ~ 0.7 during stable co-training; large oscillations may indicate imbalance.
discriminator/D(y)_prob	Mean discriminator confidence that ground-truth HR inputs are real. Should rise toward 0.8–1.0 and stay high when D is healthy.
discriminator/D(G(x))_prob	Mean discriminator confidence that generated SR outputs are real. Starts near 0 and climbs toward 0.4–0.6 as G improves realism.
generator/content_loss	Weighted content component of the generator objective (e.g., L1 or spectral loss). Dominant during pretraining; gradually decreases over time.
generator/total_loss	Full generator objective combining content and adversarial terms. Tracks content_loss early, then stabilises once the adversarial weight ramps up.
training/adv_loss_weight	Current adversarial weight applied to the generator loss. Stays at 0 during pretrain and linearly ramps to its configured maximum value.
validation/DISC_adversarial_loss	Discriminator loss on validation batches. Should roughly mirror the training curve; strong divergence may signal overfitting or instability.

Appendix C Experimental Configuration and Quantitative Results

This appendix provides detailed configurations, qualitative previews, and quantitative results for two representative experiments conducted with *OpenSR-SRGAN*. They correspond to typical workflows for RGB super-resolution on SEN2NAIP and multispectral Sentinel-2 band enhancement.

C.1 Experiment 1: $4\times$ RGB Super-Resolution on SEN2NAIP

In the first experiment, we evaluate a residual channel-attention (RCAB) generator on the SEN2NAIP dataset [33]. The model learns to map Sentinel-2 RGB-NIR patches at 10 m resolution to NAIP RGB-NIR targets at 2.5 m, corresponding to a $4\times$ upscaling factor. The generator follows an SRResNet-style backbone with RCAB blocks and moderate depth, paired with a global SRGAN-style discriminator. The training objective combines L1 content loss with a perceptual loss (LPIPS) and a relatively weak adversarial term to prioritise structural fidelity while still enhancing textures.

Training is conducted with an initial generator-only pretraining phase, during which only content and perceptual losses are active, followed by a gradual ramp-up of the adversarial weight. EMA with a high decay factor (e.g., 0.999) is used to stabilise the generator during validation. The experiment is run in distributed data-parallel mode on dual A100 GPUs with mixed-precision training. The resulting model achieves validation performance in the low–30 dB PSNR range with SSIM around 0.8 on the RGB-NIR bands, while maintaining low spectral-angle (SAM) errors and strong perceptual scores (see Tables 6 and 7 below for configuration and metrics).

Qualitative inspection of false-colour composites (Figure 2) shows sharper field boundaries, improved building shapes, and enhanced road structures compared to the bicubic upscaled LR input, with minimal spectral distortions relative to the NAIP reference. The example demonstrates how the framework can be used to train perceptually enhanced RGB-NIR super-resolution models directly from YAML configurations while tracking both physical and perceptual metrics.

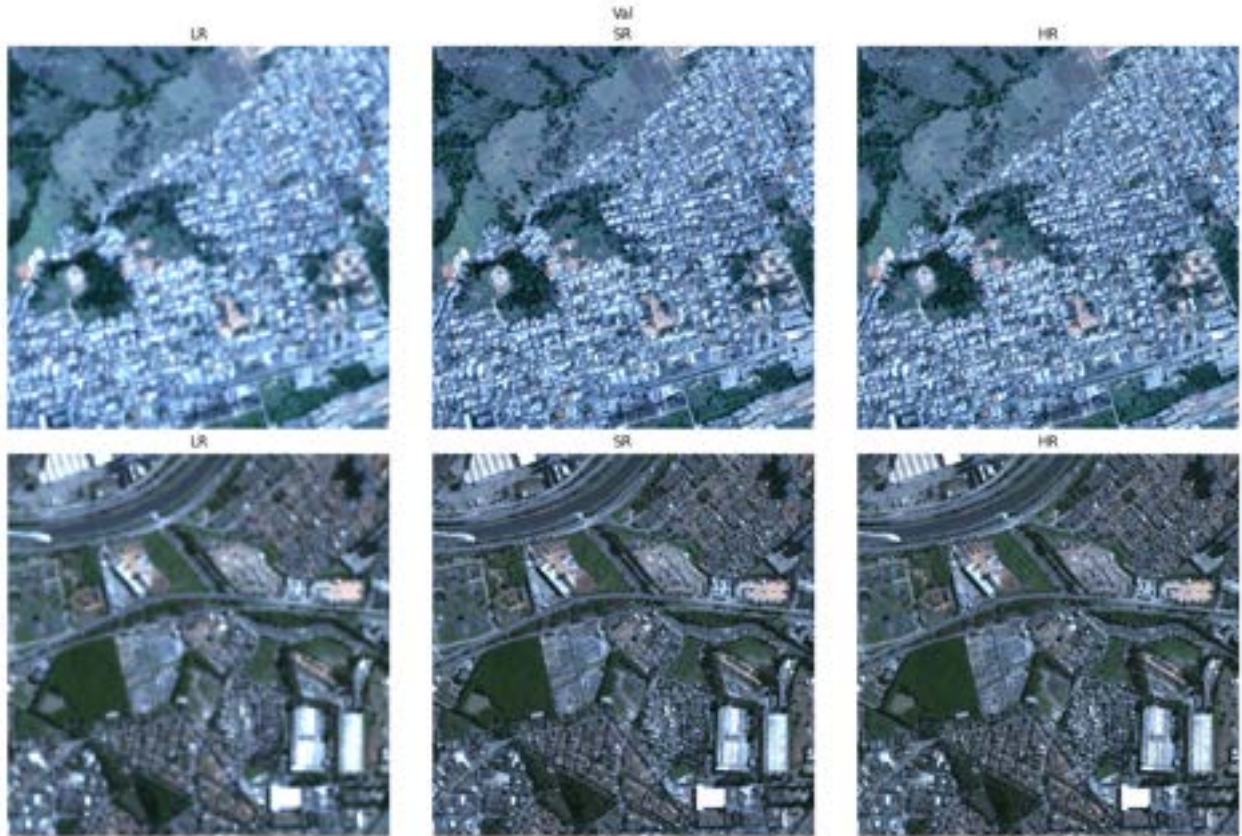


Figure 2: False-color visual comparison for $4\times$ RGB super-resolution on SEN2NAIP (Sentinel-2 input $\rightarrow$ NAIP target). Left to right: LR input, model output, HR reference.

Table 6: Configuration summary for the SEN2NAIP RGB experiment.

Parameter	Setting
Dataset	SEN2NAIP (Sentinel-2 $\rightarrow$ NAIP RGB-NIR, $4\times$ upscaling)
Generator	RCAB-based SRResNet variant (block_type=rcab, 16 blocks, 96 channels)
Discriminator	Standard global discriminator (SRGAN-style)
Loss composition	L1 (1.0) + Perceptual (0.2) + Adversarial (0.01)
Training schedule	Pretrain: 150k steps, Ramp: 50k steps, EMA $\beta = 0.999$
Hardware	Dual A100 (DDP), mixed precision (16-bit)

Table 7: Validation performance of the SEN2NAIP RGB experiment ($4\times$).

Model	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\uparrow$	SAM $\downarrow$
RCAB-SRResNet + Standard Discriminator	31.45	0.81	0.82	0.069

C.2 Experiment 2: $8\times$ 6-Band SWIR Sentinel-2 Super-Resolution

The second experiment highlights the framework’s multispectral capabilities by targeting six Sentinel-2 bands at 20 m resolution, including short-wave infrared (SWIR) channels. In the absence of native high-resolution ground truth for these bands, we adopt a pragmatic proxy setup: the original 20 m images are treated as HR references, while LR inputs are generated by downsampling to 160 m. The model is trained to recover the 20 m signal from the synthetically degraded 160 m inputs, and the learned $8\times$ upscaling operator can later be applied to the 20 m data to produce 2.5 m SR products.

Here, the generator is an SRResNet backbone with standard residual blocks, increased depth, and an upscaling factor of $8\times$. A PatchGAN discriminator focuses on local consistency and texture realism across the six spectral bands. The loss composition emphasises pixel-wise L1 reconstruction and spectral-angle (SAM) penalisation, with a relatively small adversarial weight to avoid introducing hallucinated cross-band artefacts. Compared to the first experiment, EMA is disabled to expose the raw training dynamics.

On the validation set, the model reaches mid-20 dB PSNR and SSIM around 0.7–0.75 on the reconstructed 20 m references, while achieving SAM values that indicate reasonable spectral consistency for a purely synthetic LR–HR pairing (see Tables 8 and 9 below). Figure 3 visualises three randomly selected bands, showing that the network recovers sharper edges and small structures compared to the bicubic baseline, while preserving the large-scale patterns of vegetation and built-up areas.

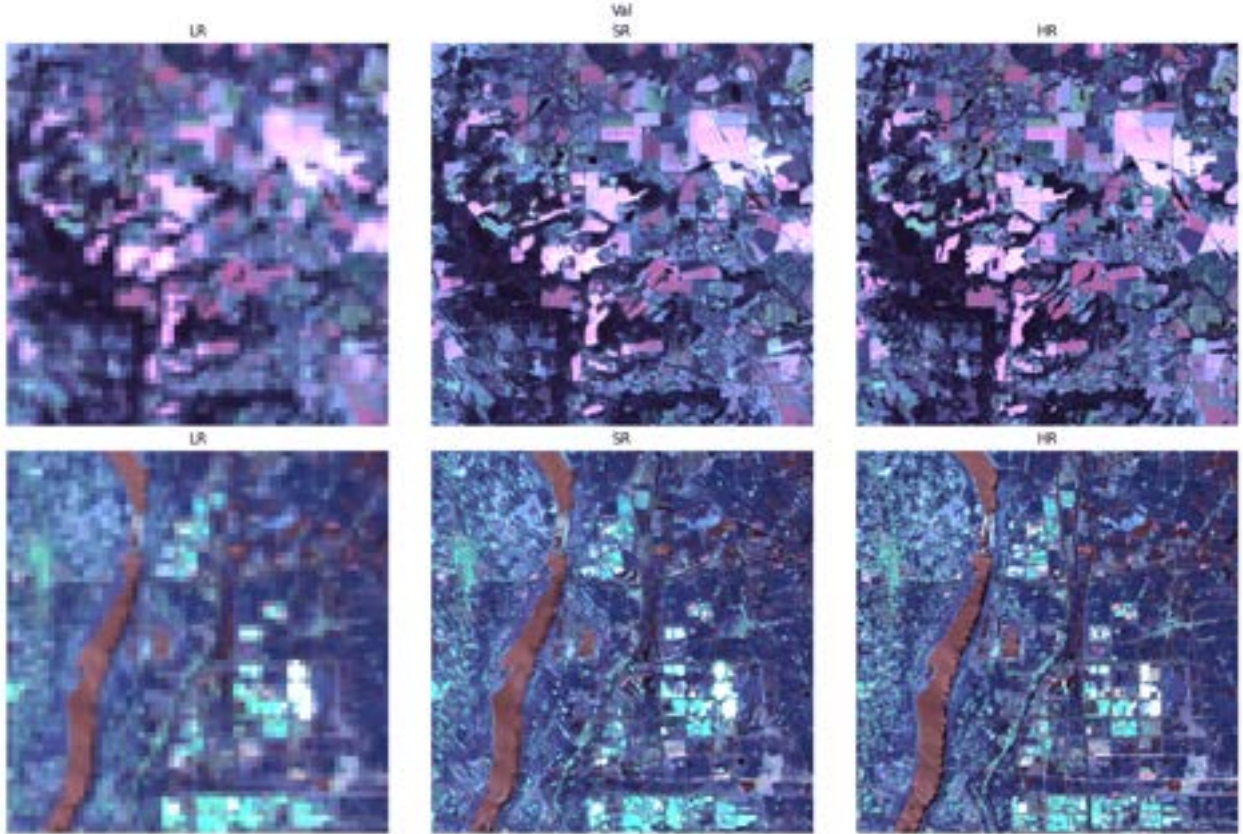


Figure 3: Visual comparison for $8\times$ multispectral super-resolution (Sentinel-2 6-band input). Left to right: LR input, model output, HR reference. Three bands are shown for visualisation.

Table 8: **Configuration summary for the 6-band Sentinel-2 experiment.**

Parameter	Setting
Dataset	Sentinel-2 6-band subset (160 m $\rightarrow$ 20 m, $8\times$ upscaling)
Generator	SRResNet backbone (block_type=res, 32 blocks, 96 channels, scale=8)
Discriminator	PatchGAN (n_blocks=4, patch size $\approx 70 \times 70$)
Loss composition	L1 (1.0) + SAM (0.2) + Adversarial (0.005)
Training schedule	Pretrain: 100k steps, Ramp: 40k steps, EMA disabled
Hardware	Dual A100 GPU, full-precision (32-bit)

Table 9: **Validation performance of the 6-band Sentinel-2 experiment ($8\times$).**

Model	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\uparrow$	SAM $\downarrow$
SRResNet (6-band) + PatchGAN Discriminator	26.65	0.74	0.80	0.091